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**Reconstruction, Energy Calibration, and Identification of
Hadronically Decaying Tau Leptons in the ATLAS Experiment for
Run-2 of the LHC**

The ATLAS Collaboration

Abstract

The reconstruction algorithm, energy calibration, and identification methods for hadronically decaying tau leptons in ATLAS used at the start of Run-2 of the Large Hadron Collider are described in this note. All algorithms have been optimised for Run-2 conditions. The energy calibration relies on Monte Carlo samples with hadronic tau lepton decays, and applies multiplicative factors based on the p_T of the reconstructed tau lepton to the energy measurements in the calorimeters. The identification employs boosted decision trees. Systematic uncertainties on the energy scale, reconstruction efficiency and identification efficiency of hadronically decaying tau leptons are determined using Monte Carlo samples that simulate varying conditions.

1 Introduction

This note describes the updates in the reconstruction, identification and energy calibration of hadronically decaying tau leptons in ATLAS [1], introduced for the beginning of the $\sqrt{s} = 13$ TeV run of the Large Hadron Collider (LHC) [2]. It summarizes the changes with respect to the methods used during the $\sqrt{s} = 8$ TeV run [3] and provides the expected performance based on simulation. Systematic uncertainties on the energy scale as well as on the reconstruction and identification efficiency are evaluated based on simulation studies and on the experience from previous data-taking periods.

Performance changes are expected mainly from the addition of a new layer of the pixel detector, the IBL [4], which improves the impact parameter and tau decay vertex resolution, as well as from the calorimeter readout configuration, updated for operation with 25 ns bunch spacing. Discrimination against electrons has been harmonized with the electron identification algorithm, instead of implementing a dedicated veto. In addition there were many updates of a technical nature, due to the developments of the ATLAS software framework.

After a description of the ATLAS detector in section 2, the simulated samples used for the optimization, determination of performance, and evaluation of the systematic uncertainty are described in section 3. Reconstruction of the hadronic component of tau decays (in the following indicated as $\tau_{\text{had-vis}}$) is outlined in section 4. The discrimination of hadronic tau decays from the main backgrounds, hadronic jets and electrons, is discussed in section 5, while section 6 presents the energy calibration and resolution.

2 ATLAS detector

The ATLAS detector [1] consists of an inner tracking system surrounded by a superconducting solenoid, electromagnetic (EM) and hadronic (HAD) calorimeters, and a muon spectrometer (MS).

The inner detector (ID) is immersed in a 2 T axial magnetic field, and consists of pixel and silicon microstrip (SCT) detectors inside a transition radiation tracker (TRT), providing charged-particle tracking in the region $|\eta| < 2.5$ ¹. For the $\sqrt{s} = 13$ TeV run, a fourth layer of the pixel detector, the IBL [4], has been inserted at an average radius of 33.2 mm, providing an additional position measurement with 8 μm resolution in the (x, y) plane and 40 μm along z .

The EM calorimeter (ECAL) uses lead as the absorber and liquid argon (LAr) as the active material. In the central rapidity region, the EM calorimeter is divided into three layers, one of them segmented in highly granular η strips for optimal γ/π^0 separation, and completed by a presampler layer for $|\eta| < 1.8$. Hadron calorimetry is based on different detector technologies, with scintillator tiles ($|\eta| < 1.7$) or LAr ($1.5 < |\eta| < 4.9$) as the active medium, and with steel, copper, or tungsten as the absorber material. The calorimeters provide coverage within $|\eta| < 4.9$.

The MS consists of superconducting air-core toroids, a system of trigger chambers covering the range $|\eta| < 2.4$, and high-precision tracking chambers allowing muon momentum measurements for $|\eta| < 2.7$.

¹ ATLAS uses a right-handed coordinate system with its origin at the nominal interaction point (IP) in the centre of the detector and the z -axis along the beam direction. The x -axis points from the IP to the centre of the LHC ring, and the y -axis points upward. Cylindrical coordinates (r, ϕ) are used in the transverse (x, y) plane, ϕ being the azimuthal angle around the beam direction. The pseudorapidity is defined in terms of the polar angle θ as $\eta = -\ln \tan(\theta/2)$. The distance ΔR in the $\eta - \phi$ space is defined as $\Delta R = \sqrt{\Delta\eta^2 + \Delta\phi^2}$.

Physics objects are identified using their specific detector signatures; electrons are reconstructed by matching a track from the ID to an energy deposit in the calorimeters [5, 6], while muons are reconstructed using tracks from the MS and ID [7]. Jets are reconstructed using the anti- k_t algorithm [8] with a distance parameter $R = 0.4$. Three-dimensional clusters of calorimeter cells, termed *TopoClusters*, calibrated using a local hadronic calibration [9], serve as inputs to the jet algorithm.

3 Event samples and sources of systematic uncertainties

The $\tau_{\text{had-vis}}$ reconstruction, identification and calibration are tuned using $Z/\gamma^* \rightarrow \tau\tau$ events simulated using Powheg-Box v2 [10] interfaced to the Pythia 8.186 [11] parton shower model. The CT10 PDF set [12] is used in the matrix element. The AZNLO [13] tune is used, with the PDF set CTEQ6L1 [14], for the modeling of non-perturbative effects. The EvtGen v1.2.0 program [15] is used for the modelling of the bottom and charm hadron decays. Photos++ version 3.52 [16] is used for QED emissions from electroweak vertices and charged leptons. Generated events are processed through the ATLAS detector simulation [17] based on GEANT4 [18]. The Monte Carlo (MC) simulation samples consist of an inclusive $Z/\gamma^* \rightarrow \tau\tau$ sample, while high-mass Drell-Yan $\tau^+\tau^-$ samples cover the high- p_T part of the spectrum.

The identification algorithms are optimized using background samples of di-jet events, simulated using Pythia 8.175 [11] with AU2 tune [19] and CT10 PDF [12], and of $Z/\gamma^* \rightarrow ee$ events with the same generator settings as for the inclusive $Z/\gamma^* \rightarrow \tau\tau$ sample.

The nominal samples are produced with the expected ATLAS configuration and conditions for the 13 TeV operation with 25 ns bunch spacing. The pileup due to multiple pp interactions per bunch crossing is simulated by adding event-by-event a number of minimum bias interactions extracted from a Poisson distribution with average number of interaction per bunch crossing μ . The value of μ follows a distribution covering the range of values expected during the 2015 data-taking, with a mean value about 25. The interactions of particles with the detector are simulated by GEANT4 using the FTFP_BERT hadronic shower model.

The following systematic uncertainties on the performance have been evaluated using dedicated $Z/\gamma^* \rightarrow \tau\tau$ samples generated with different configurations:

- Uncertainty on the inner detector material, which affects track reconstruction efficiency, photon conversion rate and the amount of energy loss in front of the calorimeter, has been studied by increasing the passive material in the inner detector by 5%. A more precise modeling of the detector material, using early data at $\sqrt{s} = 13$ TeV, has been used in recent publications [20, 21], but it has been checked it doesn't have a sizeable effect on the uncertainty quoted on this systematic.
- Uncertainty in the calorimeter performance and calibration have been estimated by varying the noise thresholds in the reconstruction of *TopoClusters*. Both upwards and downwards variations have been considered and the absolute value of the maximum deviation is quoted as the systematic uncertainty.
- Uncertainty due to underlying event tunes are derived by comparing the nominal sample with a sample simulated using the A14 tune [22].
- Events generated with the QGSP_BIC GEANT4 hadronic shower model are used to evaluate uncertainties on the modeling of interaction with the detector.

In addition, uncertainties due to the pileup mismodeling have been evaluated by dividing the nominal sample into four μ bins and taking half of the difference between the uppermost and lowermost bin.

Systematic uncertainties are evaluated in different $|\eta|$ regions as a function of the $\tau_{\text{had-vis}}$ candidate p_T , after the calibration described in section 6.

4 Reconstruction

The $\tau_{\text{had-vis}}$ reconstruction algorithm used in the 2012 data-taking has been deployed with minor modification for the 13 TeV data-taking. It is described in Ref. [3] and summarized in the following. Jets formed using the anti- k_t algorithm with a distance parameter $R = 0.4$ and clusters of calorimeter cells, calibrated using a local hadronic calibration (LC) as inputs, are used as seeds of the $\tau_{\text{had-vis}}$ reconstruction algorithm if they satisfy $p_T > 10$ GeV and $|\eta| < 2.5$.

The $\tau_{\text{had-vis}}$ p_T at the LC scale is set to the total energy of TopoClusters within $\Delta R < 0.2$. This is the energy scale used in the identification variables of section 5.1. This is also the base value for the final energy calibration discussed in section 6.1.

In $\tau_{\text{had-vis}}$ reconstruction, the hadronic component (HAD) of the TopoClusters consists of the cells in the last layer of the electromagnetic calorimeter and of the hadronic calorimeter, while the electromagnetic (EM) component is referred to the energy deposited in the other layers of the ECAL.

The performance of the efficiencies and calibration in this note are evaluated for $\tau_{\text{had-vis}}$ candidates with $|\eta| < 2.5$, outside the transition region $1.37 < |\eta| < 1.52$ between the barrel and forward calorimeters, and with a reconstructed $p_T > 20$ GeV, after the calibration described in section 6.

4.1 Vertex association

In events with multiple simultaneous interactions, the chosen primary vertex does not always correspond to the vertex at which the tau lepton is produced. To reduce the effects of pile-up and increase reconstruction efficiency, the tau lepton production vertex is identified, among the previously reconstructed primary vertex candidates in the event.

The tau vertex (TV) association algorithm uses as input all tau candidate tracks in the region $\Delta R < 0.2$ around the jet seed direction. The p_T of these tracks is summed and the primary vertex candidate to which the largest fraction of the p_T sum is matched is chosen as the TV [23].

New impact parameters can then be recalculated with respect to the TV and $\tau_{\text{had-vis}}$ tracks are selected using the criteria described in Section 4.2. This vertex is used to determine the $\tau_{\text{had-vis}}$ direction, to associate tracks, and to build the coordinate system in which identification variables presented in section 5 are calculated.

In Fig. 1, the TV performance is compared to the general purpose algorithm used in ATLAS, which selects the primary vertex that maximize the sum of p_T^2 of associated tracks. The TV provides a better efficiency for the reconstruction of low- p_T $\tau_{\text{had-vis}}$ candidates, especially in a high pile-up environment.

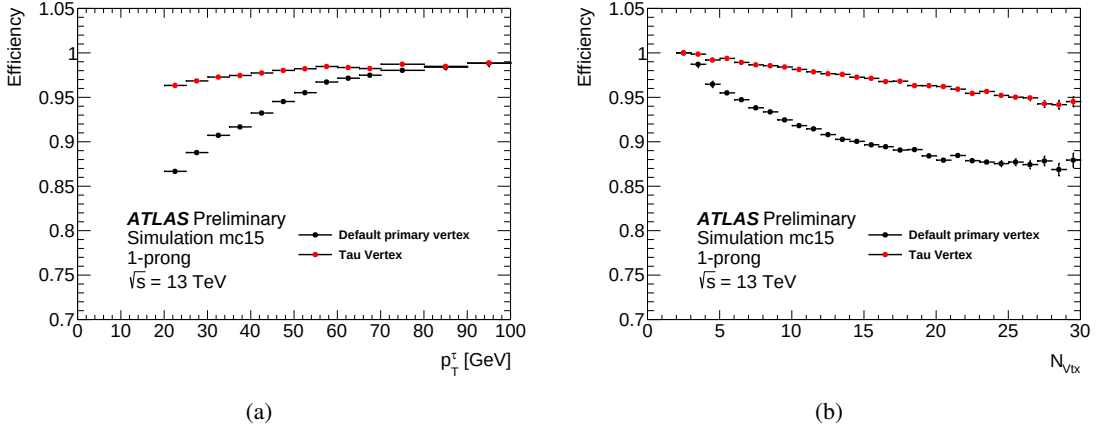


Figure 1: Efficiency for correct production vertex assignment in 1-prong tau decays for the tau reconstruction algorithm and the default choice of the vertex with the highest $\sum p_T^2$, as a function of $\tau_{had-vis}$ p_T (a) and of the number of reconstructed vertices in the event (b).

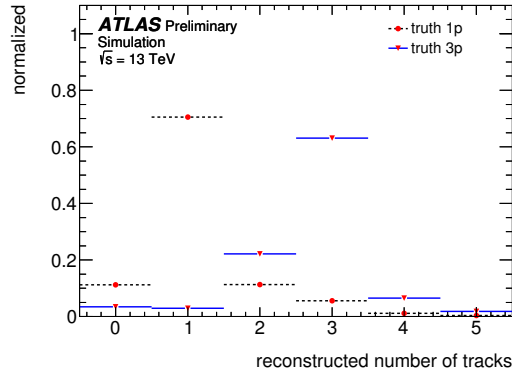


Figure 2: Number of reconstructed tracks for $\tau_{had-vis}$ candidates from true 1-prong and 3-prong tau decays.

4.2 Track selection

Tracks are associated to the $\tau_{had-vis}$ candidate following the same criteria used in Run-1. They are required to be in the core region $\Delta R < 0.2$ around the $\tau_{had-vis}$ direction and to satisfy the following criteria: $p_T > 1$ GeV, at least two associated hits in the pixel detector (including the IBL), and at least seven hits in total in the pixel and the SCT detectors. Furthermore, requirements are imposed on the distance of closest approach of the track to the TV in the transverse plane, $|d_0| < 1.0$ mm, and longitudinally, $|\Delta z_0 \sin \theta| < 1.5$ mm. This selection is optimized to maximize the fraction of 1-prong and 3-prong tau decays reconstructed with the correct charged particle multiplicity.

The distribution of the number of tracks associated to $\tau_{had-vis}$ candidates, separately for 1-prong and 3-prong decays, is displayed in Fig. 2. The leading cause of underestimation of the number of prongs is tracking inefficiency due to hadronic interactions in the inner detector, while the leading cause for overestimation of the number of prongs comes from photon conversion tracks that pass the track selection criteria.

4.3 Reconstruction efficiency and systematic uncertainties

The reconstruction efficiency is defined as the fraction of 1-prong (3-prong) hadronic tau decays which are reconstructed as 1-track (3-track) $\tau_{\text{had-vis}}$ candidates.

The detector is almost fully efficient for finding a jet-seed for a $\tau_{\text{had-vis}}$ candidate with $p_T > 20$ GeV within the pseudorapidity acceptance. Track and vertex selections, described in sections 4.1 and 4.2 therefore play the dominant role for the reconstruction efficiency shown in Fig. 3. The reconstruction efficiency is relatively constant for 1-prong decays, with a slow drop at high momentum due to two effects. Very high- p_T tau leptons may decay after the first pixel detector and fail the requirement on the number of hits. In addition, the probability to wrongly classify an electron from photon conversions as a charged hadron from tau decays also increases with p_T and results in an incorrect assignment of the number of charged particles in the tau decay. For 3-prong decays there is a reduction in the low- p_T bins, due to the minimum transverse momentum requirement on the charged decay products, and at high- p_T due to the increased collimation of the decay products that results in an increased probability to miss a track because of overlapping trajectories.

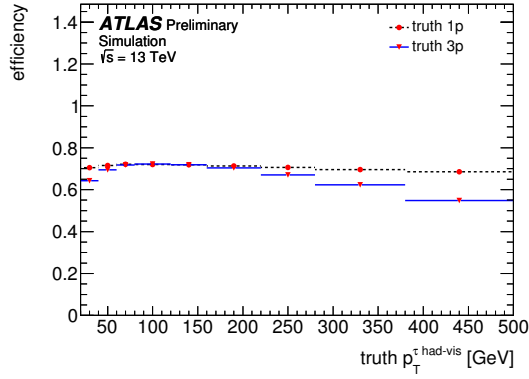


Figure 3: Efficiency for reconstructing the same number of tracks as the number of charged decay products of the tau lepton as a function of $\tau_{\text{had-vis}}$ p_T .

The relative systematic uncertainties on the reconstruction efficiency are summarized in Fig. 4. The effect of the systematics variations is found to be uniform along η and the sum of all contributions is in the range between 2% and 4.5%. The main sources of systematic uncertainties are from pileup and the inner detector material. The former affects the efficiency mainly through the TV, where the probability to incorrectly identify the $\tau_{\text{had-vis}}$ candidate production vertex depends on the amount of interactions reconstructed in the event, as shown in Fig. 1(b), and is especially relevant for low- p_T 1-prong decays. The latter impacts directly the track counting, as the number of reconstructed conversion tracks and the efficiency to reconstruct pion tracks can both be affected by the amount of detector material.

5 Identification

The reconstruction of $\tau_{\text{had-vis}}$ candidates provides very little rejection against the jet background. Rejection against jets is provided in a separate identification step using the discriminating variables described in section 5.1, which are based on the tracks and TopoClusters (and cells linked to them) found in the

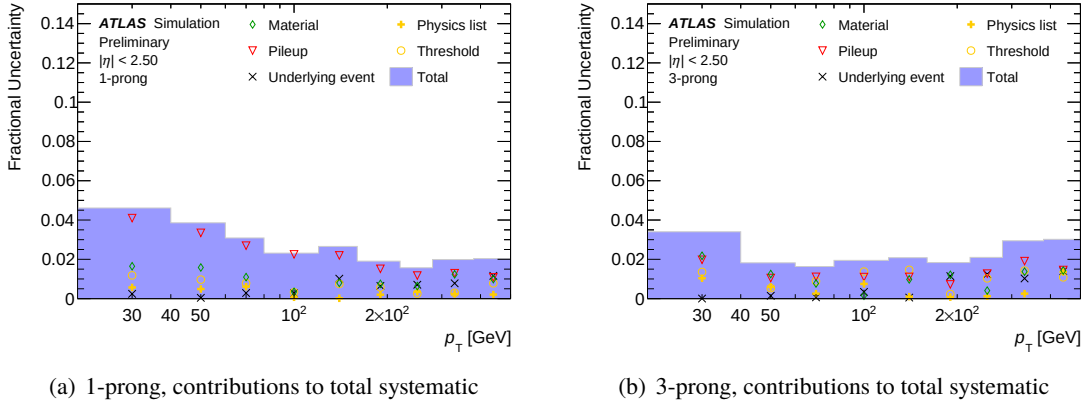


Figure 4: Relative systematic uncertainties on the efficiency for reconstructing the same number of tracks as the number of charged decay products of the tau lepton as a function of $\tau_{\text{had-vis}} p_T$. The different sources of systematic uncertainties are listed in the legend. The individual contributions to the total systematic are shown for true 1-prong decays in (a) and for true 3-prong decays in (b).

core or isolation region around the $\tau_{\text{had-vis}}$ candidate direction. The calorimeter measurements provide information about the longitudinal and lateral shower shape. The ECAL is sensitive to the π^0 content of tau hadronic decays.

The identification of $\tau_{\text{had-vis}}$ candidates to discriminate tau lepton decays from hadronic jets follows the same general approach used in the 2012 data-taking and is described in Ref. [3]. Separate Boosted Decision Trees (BDT) algorithms are trained for 1-track and 3-track $\tau_{\text{had-vis}}$ decays using the samples listed in section 3: $Z/\gamma^* \rightarrow \tau\tau$ for signal and di-jet events for background.

One notable difference compared to the 2012 data-taking is that the identification variables and pile up correction procedure are now uniform between offline reconstruction and trigger. Part of this harmonization was the replacement of the π^0 reconstruction algorithm described in section 3.1 of Ref. [3] with some of its input variables and up to two EM TopoClusters for the π^0 component. This variable choice provides an equivalent level of rejection compared to the original algorithm.

5.1 Identification variables

The full list of discriminating variables used for tau identification is given below and is summarized in Table 1. Most of the variables are the same as those used in the 2012 data-taking [3]:

Central energy fraction (f_{cent}): Fraction of the calorimeter transverse energy deposited in the region $\Delta R < 0.1$ with respect to all energy deposited in the region $\Delta R < 0.2$ around the $\tau_{\text{had-vis}}$ candidate. It is calculated by summing the energy deposited in all cells belonging to TopoClusters with a barycentre in these regions, calibrated at the EM energy scale.

Leading track momentum fraction ($f_{\text{leadtrack}}^{-1}$): The transverse energy sum, calibrated at the EM energy scale, deposited in all cells belonging to TopoClusters in the core region of the $\tau_{\text{had-vis}}$ candidate, divided by the transverse momentum of the highest- p_T charged particle in the core region.

- Track radius ($R_{\text{track}}^{0.2}$):** p_T -weighted ΔR distance of the associated tracks to the $\tau_{\text{had-vis}}$ direction, using only tracks in the core region.
- Leading track IP significance ($|S_{\text{leadtrack}}|$):** Absolute value of transverse impact parameter of the highest- p_T track in the core region, calculated with respect to the TV, divided by its estimated uncertainty.
- Fraction of tracks p_T in the isolation region ($f_{\text{iso}}^{\text{track}}$):** Scalar sum of the p_T of tracks associated with the $\tau_{\text{had-vis}}$ candidate in the region $0.2 < \Delta R < 0.4$ divided by the sum of the p_T of all tracks associated with the $\tau_{\text{had-vis}}$ candidate..
- Maximum ΔR (ΔR_{Max}):** The maximum ΔR between a track associated with the $\tau_{\text{had-vis}}$ candidate and the $\tau_{\text{had-vis}}$ direction. Only tracks in the core region are considered.
- Transverse flight path significance (S_T^{flight}):** The decay length of the secondary vertex (vertex reconstructed from the tracks associated with the core region of the $\tau_{\text{had-vis}}$ candidate) in the transverse plane, calculated with respect to the TV, divided by its estimated uncertainty. It is defined only for multi-track $\tau_{\text{had-vis}}$ candidates.
- Track mass (m_{track}):** Invariant mass calculated from the sum of the four-momentum of all tracks in the core and isolation regions, assuming a pion mass for each track.
- Fraction of EM energy from charged pions ($f_{\text{EM}}^{\text{track-HAD}}$):** Fraction of the electromagnetic energy of tracks associated with the $\tau_{\text{had-vis}}$ candidate in the core region. The numerator is defined as difference between the sum of the momentum of tracks in the core region and the sum of cluster energy deposited in the hadronic part of each TopoCluster (including the third layer of the EM calorimeter) associated with the $\tau_{\text{had-vis}}$ candidate. The denominator is the sum of cluster energy deposited in the electromagnetic part of each TopoCluster (presampler and first two layers of the EM calorimeter) associated with the $\tau_{\text{had-vis}}$ candidate. All clusters are calibrated at the LC energy scale.
- Ratio of EM energy to track momentum ($f_{\text{track}}^{\text{EM}}$):** Ratio of the sum of cluster energy deposited in the electromagnetic part of each TopoCluster associated with the $\tau_{\text{had-vis}}$ candidate to the sum of the momentum of tracks in the core region. All clusters are calibrated at the LC energy scale.
- Track-plus-EM-system mass ($m_{\text{EM+track}}$):** Invariant mass of the system composed of the tracks and up to two most energetic EM clusters in the core region, where EM cluster energy is the part of TopoCluster energy deposited in the presampler and first two layers of the EM calorimeter, and the four-momentum of an EM cluster is calculated assuming zero mass and using TopoCluster seed direction.
- Ratio of track-plus-EM-system to p_T ($p_T^{\text{EM+track}}/p_T$):** Ratio of the $\tau_{\text{had-vis}}$ p_T , estimated using the vector sum of track momenta and up to two most energetic EM clusters in the core region to the calorimeter-only measurement of $\tau_{\text{had-vis}}$ p_T .

A correction depending linearly on μ , the average number of pileup interactions per bunch crossing computed from the instantaneous luminosity [24], is applied to each discriminating variable. The usage of μ , instead of the number of reconstructed interaction vertices per event, N_{PV} , provides compatibility with the High Level Trigger, that cannot run a full event primary vertex fit.

Compared to the identification variables used in Run-1, the current set presents the following updates:

- The variable $f_{\text{iso}}^{\text{track}}$ replaces $N_{\text{track}}^{\text{iso}}$, being more consistent with other relative isolation variables used in ATLAS. The distribution of this variable for signal and background events is shown in Fig. 5.
- The track radius cone size was reduced from 0.4 to 0.2 to be more pileup robust.
- The leading track IP significance is redefined. The new definition avoids the need by the BDT to apply symmetric cuts and also allows the application of the pileup correction procedure.
- The variable $f_{\text{leadtrack}}^{-1}$ is inverse to the f_{track} variable used in the 2012 data-taking, which is motivated by Run-2 event data model (EDM).
- The variables $f_{\text{track}}^{\text{EM}}$ and $f_{\text{EM}}^{\text{track-HAD}}$ are input variables to the π^0 counting algorithm used for 2012 data. The $m_{\text{EM+track}}$ and $p_{\text{T}}^{\text{EM+track}}/p_{\text{T}}$ approximate the variables $m_{\pi^0+\text{track}}$ and $p_{\text{T}}^{\pi^0+\text{track}}/p_{\text{T}}$ computed by this algorithm, using the most energetic EM clusters. This set of variables provides the same discriminating power as the original set. Distribution of these variables for signal and background $\tau_{\text{had-vis}}$ candidates is shown in Fig. 6.

The $|S_{\text{leadtrack}}|$ and $S_{\text{T}}^{\text{flight}}$ benefit from the improved tracking resolution that results from the installation of the IBL [4]. The distributions of these variables for signal and background $\tau_{\text{had-vis}}$ candidates are shown in Fig. 7.

Variable	Offline	
	1-track	3-track
f_{cent}	•	•
$f_{\text{leadtrack}}^{-1}$	•	•
$R_{\text{track}}^{0.2}$	•	•
$ S_{\text{leadtrack}} $	•	
$f_{\text{iso}}^{\text{track}}$	•	
ΔR_{Max}		•
$S_{\text{T}}^{\text{flight}}$		•
m_{track}		•
$f_{\text{EM}}^{\text{track-HAD}}$	•	•
$f_{\text{track}}^{\text{EM}}$	•	•
$m_{\text{EM+track}}$	•	•
$p_{\text{T}}^{\text{EM+track}}/p_{\text{T}}$	•	•

Table 1: Discriminating variables used as input to the tau identification algorithm at offline reconstruction and at trigger level, for 1-track and 3-track candidates. The bullets indicate whether a particular variable is used for a given selection. The $|S_{\text{leadtrack}}|$ and $S_{\text{T}}^{\text{flight}}$ variables are calculated with respect to beam line in the trigger.

5.2 Performance and systematics

The BDT score distributions for signal and background events is displayed in Fig. 8(a) for 1-prong decays and in Fig. 8(b) for 3-prong ones. The performance of the tau identification algorithm in terms of the inverse background efficiency, $1/\varepsilon_{\text{bkg}}$ versus the signal identification efficiency, ε_{sig} is shown in Fig. 9. The identification efficiency is defined as the fraction of 1-prong (3-prong) hadronic tau decays that are

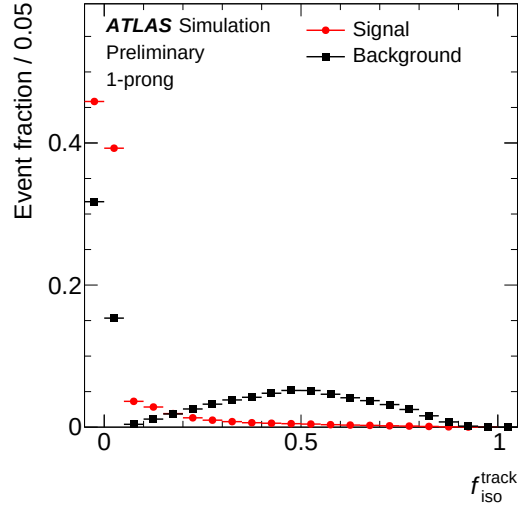


Figure 5: Distribution of the $f_{\text{iso}}^{\text{track}}$ isolation variable for hadronic tau decays (red circles) and simulated multi-jet events (black squares).

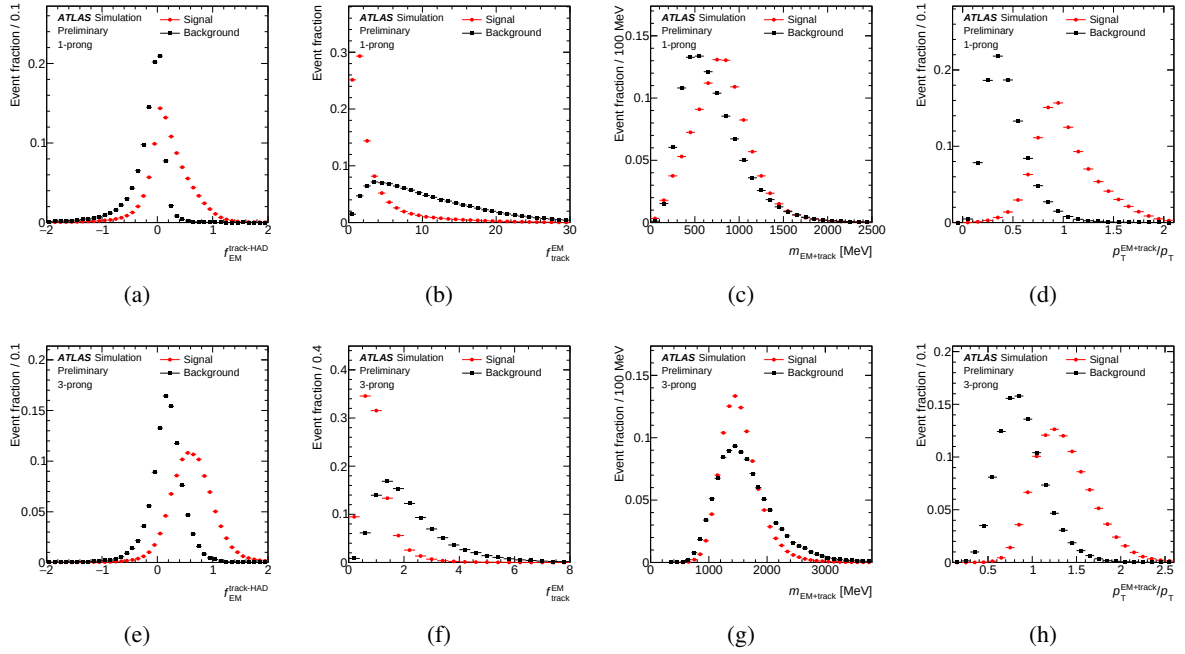


Figure 6: Distribution of the new identification variables for tau leptons (red circles) and multi-jet background (black squares) for 1-prong (top) and 3-prong (bottom). From left to right: $f_{\text{EM}}^{\text{track-HAD}}$, $f_{\text{track}}^{\text{EM}}$, $m_{\text{EM+track}}$ and $p_{\text{T}}^{\text{EM+track}}/p_{\text{T}}$.

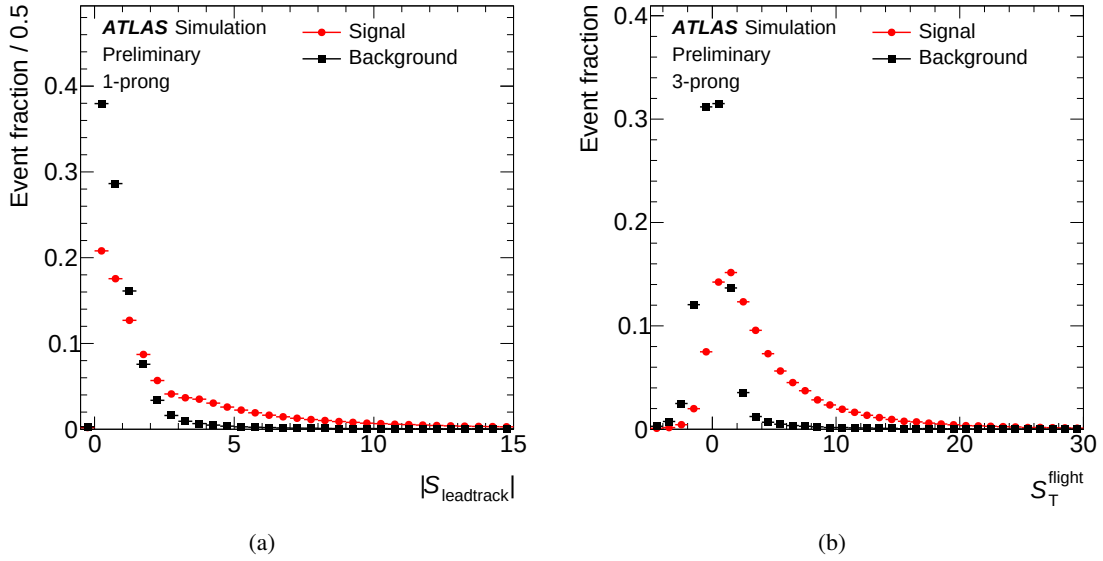


Figure 7: Distribution of the lifetime sensitive variables for hadronic τ decays (red circles) and simulated multi-jet events (black squares): the impact parameter significance $|S_{\text{leadtrack}}|$ for 1-track candidates (a) and the significance of the transverse flight path S_T^{flight} for 3-track candidates (b).

reconstructed as 1-track (3-track) $\tau_{\text{had-vis}}$ candidates, which also pass the BDT selection criteria. The efficiency is the product of the reconstruction efficiency defined in section 4.3 and of the identification efficiency.

Three working points, labelled *Tight*, *Medium* and *Loose*, are provided, corresponding to different tau identification efficiency values. For each working point, requirements on the BDT score are determined as a function of $\tau_{\text{had-vis}}$ p_T , in order to achieve a constant value for the combined reconstruction and identification signal efficiency, where the reconstruction efficiency is defined in section 4.3 and the identification efficiency is defined with respect to correctly reconstructed 1-track or 3-track $\tau_{\text{had-vis}}$ candidates. The performance of these working points is superimposed on Fig. 9. They do not exactly overlap the $1/\varepsilon_{\text{bkg}}$ curve because this curve is derived for constant BDT threshold requirement, and not for constant efficiency as a function of p_T .

After the tuning of the p_T -dependent BDT threshold values and the pileup corrections, the dependence of the combined reconstruction and identification efficiency as a function of p_T and μ is significantly reduced. The residual dependencies are shown in Fig. 10 for p_T and Fig. 11 for μ . The 3-prong identification efficiency shows a dip between in the 50 – 100 GeV range. This compensates the p_T -dependence of the reconstruction efficiency (Fig. 3), providing a more uniform combined efficiency. Signal efficiencies are, respectively, 0.6, 0.55 and 0.45 for the 1-prong *Loose*, *Medium* and *Tight* working points. The corresponding 3-prong working points have efficiencies of 0.5, 0.4 and 0.3.

Systematic uncertainties on the identification efficiency are evaluated as described in section 3. The effects of the variations are uniform in the central $|\eta| < 1.37$ and in the forward $|\eta| > 1.52$ regions. As an example, the resulting uncertainties for the *Medium* working point are illustrated in Fig. 12.

The largest contribution to the systematics uncertainty for $p_T < 100$ GeV is the calorimeter calibration and performance, which causes variation in shapes of calorimeter based and tracking-based input variables.

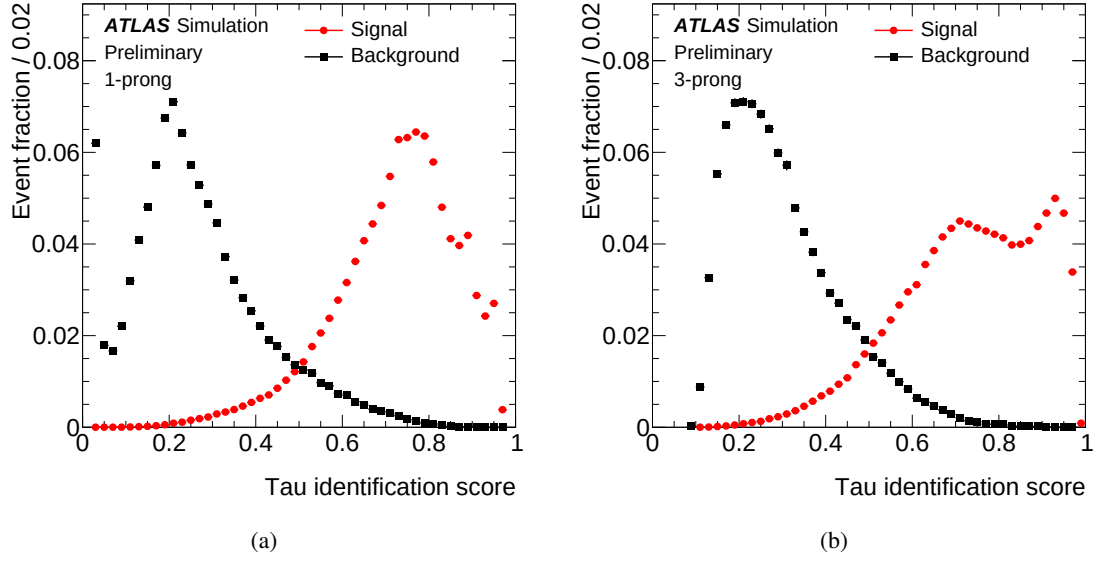


Figure 8: BDT score for hadronic tau decays (red circles) and simulated multi-jet events (black squares) for 1-track (a) and 3-track (b) $\tau_{\text{had-vis}}$ candidates.

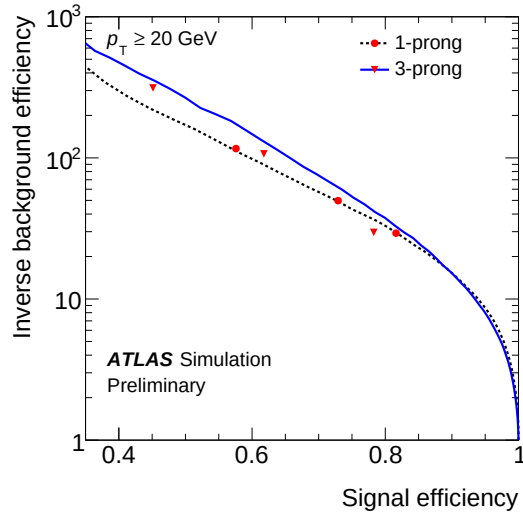


Figure 9: Inverse of the efficiency for mis-tagging QCD jets as a function of the identification efficiency for $\tau_{\text{had-vis}}$ candidates. The two lines refer to 1-track and 3-track candidates. The *Loose*, *Medium* and *Tight* working points are shown on these lines. The working points do not stay exactly on the line because they implement variable cuts to achieve a reduced p_T -dependency of the efficiency.

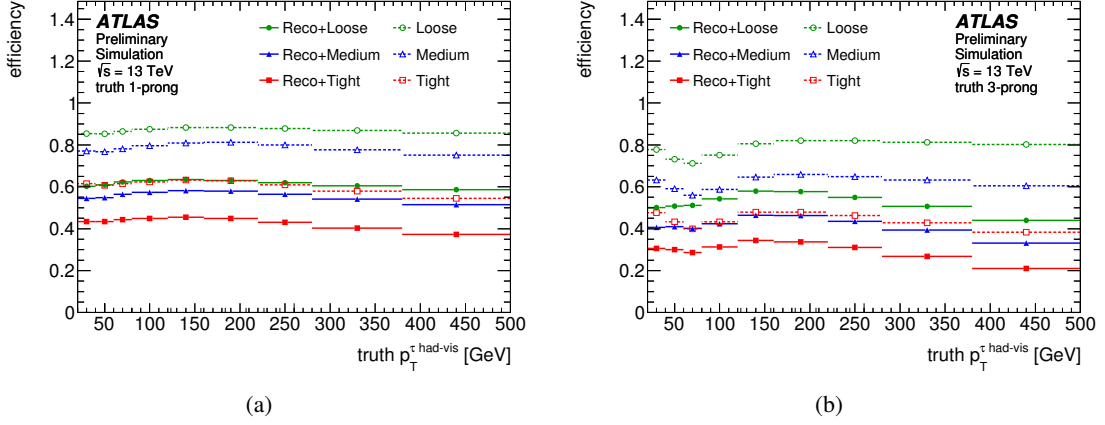


Figure 10: Efficiency for $\tau_{\text{had-vis}}$ identification (open symbols) and combined reconstruction and identification efficiency (full symbols) as a function of the $\tau_{\text{had-vis}}$ p_T , for 1-track (a) and 3-track (b) $\tau_{\text{had-vis}}$ candidates.

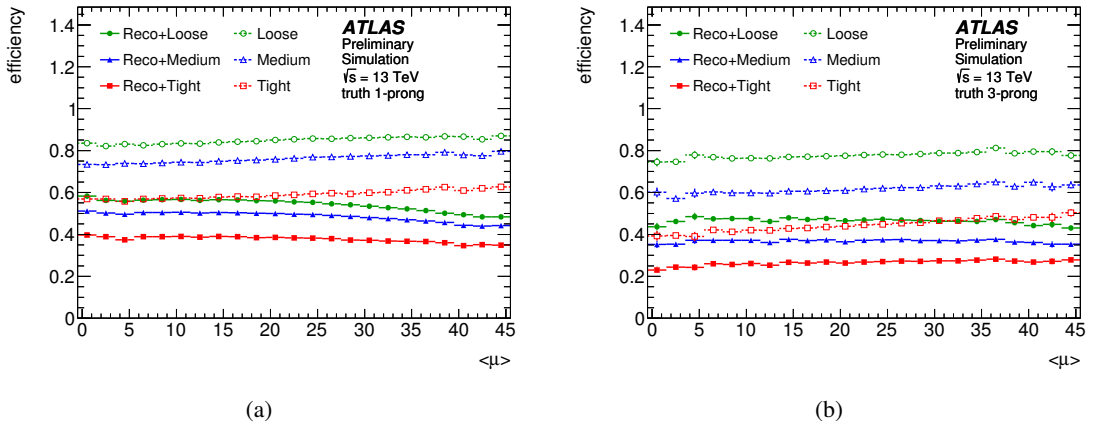


Figure 11: Efficiency for $\tau_{\text{had-vis}}$ identification (open symbols) and combined reconstruction and identification efficiency (full symbols) as a function of the average number of interactions per event, for 1-track (a) and 3-track (b) $\tau_{\text{had-vis}}$ candidates.

It reaches up to 16% for 3-prong decays in the barrel region. At large p_T , however, the pileup uncertainty dominates. For 1-prong decays the material uncertainty also plays an important role. This is mainly due the effect of the additional material on the impact parameter resolution that affects the $|S_{\text{leadtrack}}|$ distribution.

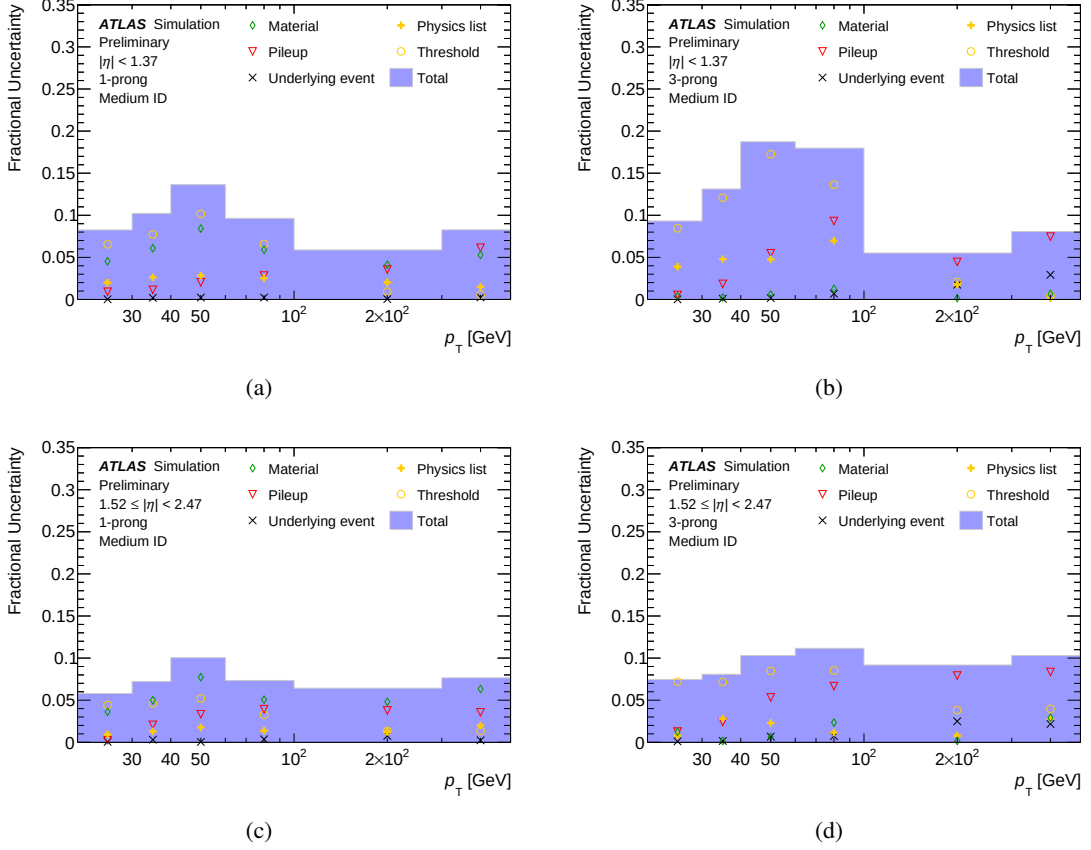


Figure 12: Relative systematic uncertainty on the efficiency for the *Medium* working point of $\tau_{\text{had-vis}}$ identification as a function of the $\tau_{\text{had-vis}}$ p_T , for 1-track (a) and 3-track (b) $\tau_{\text{had-vis}}$ candidates in the central $|\eta| < 1.37$ region, and for 1-track (c) and 3-track (d) $\tau_{\text{had-vis}}$ candidates in the forward $|\eta| > 1.52$ region. The different sources of systematic uncertainties are listed in the legend, the grey area is the square sum of the individual contribution.

5.3 Discrimination against electrons

While the BDT classifier discussed in section 5.2 was trained in order to distinguish hadronic tau decays from hadronic jets, it provides insufficient discriminating power between electrons and 1-prong hadronic tau decays. As charged pions provide one of the major backgrounds to the reconstruction of electrons, a powerful Likelihood discriminator between the two has been built in the context of the ATLAS electron reconstruction and identification effort [25]. This discriminator is based on both shower shape information from the calorimeter measurements as well as on reconstructed hits in the tracking detector, including information from transition radiation in the TRT.

In order to exploit this Likelihood discriminator for the discrimination between hadronic tau decays and electrons, a geometrical matching within a cone of $\Delta R < 0.4$ is performed between reconstructed tau

candidates and electrons with $p_T > 5$ GeV. Such a matching is necessary because there is no one-to-one correspondence between tau candidates and electron candidates. Reconstructed 1-prong tau candidates which are matched to an electron candidate of a large Likelihood score value are rejected. The cut values on the electron Likelihood score are parametrised as a function of η^{trk} and p_T to offer a constant tau efficiency of 95% with respect to a sample of tau candidates passing the *Loose* identification working point.

During electron reconstruction, loose requirements are applied on basic shower shape observables such as the hadronic leakage of energy deposits in the electromagnetic into the hadronic calorimeter. Due to these requirements, not all reconstructed tau candidates can be matched to electron candidates and are assigned an artificial Likelihood value of -4, corresponding to a very tau-like classification.

Figure 13(a) shows the distribution of the Likelihood score for a sample of true 1-prong hadronic tau decays and a sample of true electrons from $Z \rightarrow \tau\tau$ and $Z \rightarrow ee$ decays, respectively. Figure 13(b) shows the discriminating power of this classification algorithm in terms of the inverse background efficiency as a function of signal efficiency in different bins of η^{trk} .

As the electron Likelihood discriminator was optimised to provide stable performance in differing pileup scenarios, this feature is directly inherited to the electron rejection algorithm in the context of hadronic tau identification. Figures 14(a) and 14(b) show the signal efficiency as a function of p_T and μ , respectively: both are within 1% of the working point in high-statistics bins.

In order to estimate the systematic uncertainty on the signal efficiency of this procedure, the efficiency is compared between the different, systematically varied simulated event samples of section 3 and the nominal simulation. The effect of the individual variation and the total uncertainty obtained from their sum in quadrature are presented in Fig. 15. For $p_T < 100$ GeV, uncertainties are below 2%, but increase at higher p_T reaching a value of 2.5 – 5% depending on the pseudorapidity.

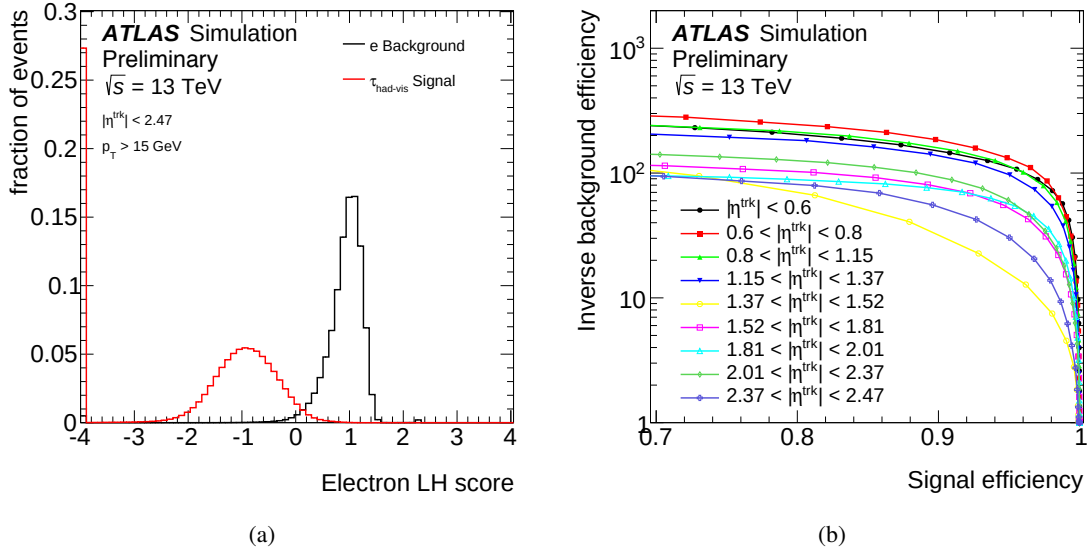


Figure 13: Electron likelihood score for hadronic tau decays and prompt electrons from $Z \rightarrow ee$ (a) and inverse of electron mis-identification efficiency as a function of the tau identification efficiency for different $|\eta|$ regions (b). The nominal working point corresponds to a tau identification efficiency of 95%.

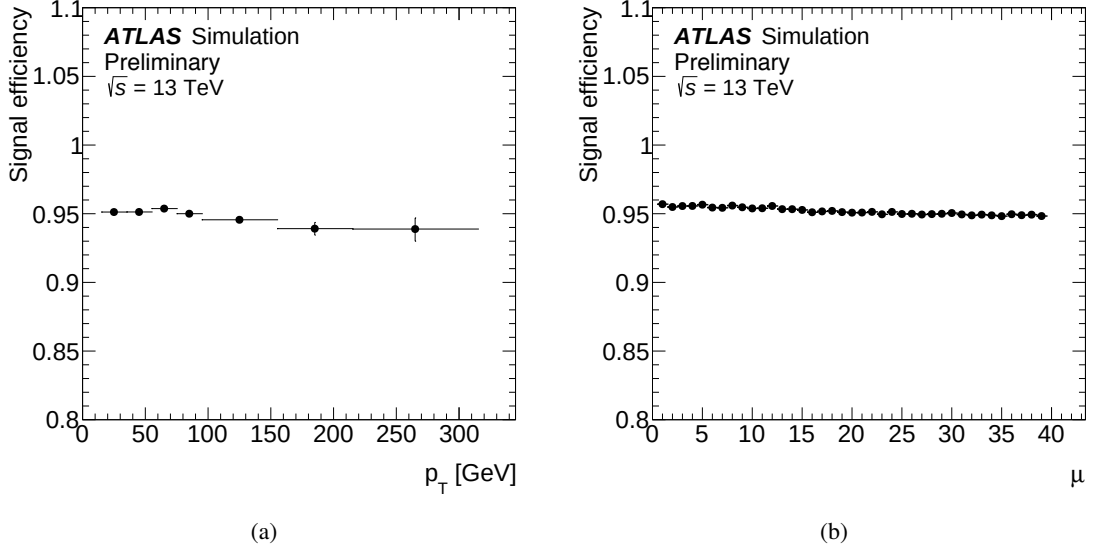


Figure 14: Efficiency of the τ - e discrimination as a function of the $\tau_{\text{had-vis}}$ p_T (a) and of the average number of pileup interactions (b). The nominal working point corresponds to a tau identification efficiency of 95%.

6 Energy calibration

As mentioned in section 4, the tau energy scale calibration developed for the start of Run-2 uses calorimeter information only. After the reconstruction step, the energy of tau candidates is calibrated at the LC scale, which mostly corrects for the calorimeter non-compensation and for the energy deposited in dead material or outside TopoClusters.

The tau energy scale consists of two additional corrections which calibrate the tau energy back to the true visible energy. First, the energy contribution originating from pileup interactions is subtracted. A response correction is then applied to account for the following possible effects for the $\tau_{\text{had-vis}}$ particles: decay products not reaching the calorimeter, not depositing enough energy to create TopoClusters, or not detected within $\Delta R = 0.2$ of the reconstructed $\tau_{\text{had-vis}}$ candidate. The $\tau_{\text{had-vis}}$ momentum is computed from the energy assuming zero mass.

6.1 Calibration functions

The calibrated tau energy is computed as

$$E_{\text{calib}} = \frac{E_{\text{LC}} - E_{\text{pileup}}}{\mathcal{R}(E_{\text{LC}} - E_{\text{pileup}}, |\eta|, n_p)}, \quad (1)$$

where $E_{\text{pile-up}}$ is the pile-up offset correction, $\mathcal{R}$ is the calibration function also called response in the following, and n_p refers to 1-prong or multi-prong decays. The LC-calibrated energy E_{LC} is found to increase linearly with the number of primary vertices N_{PV} . Thus, the pile-up correction is derived as

$$E_{\text{pileup}}(N_{\text{PV}}, |\eta|, n_p) = A(|\eta|, n_p) \times (N_{\text{PV}} - \langle N_{\text{PV}} \rangle). \quad (2)$$

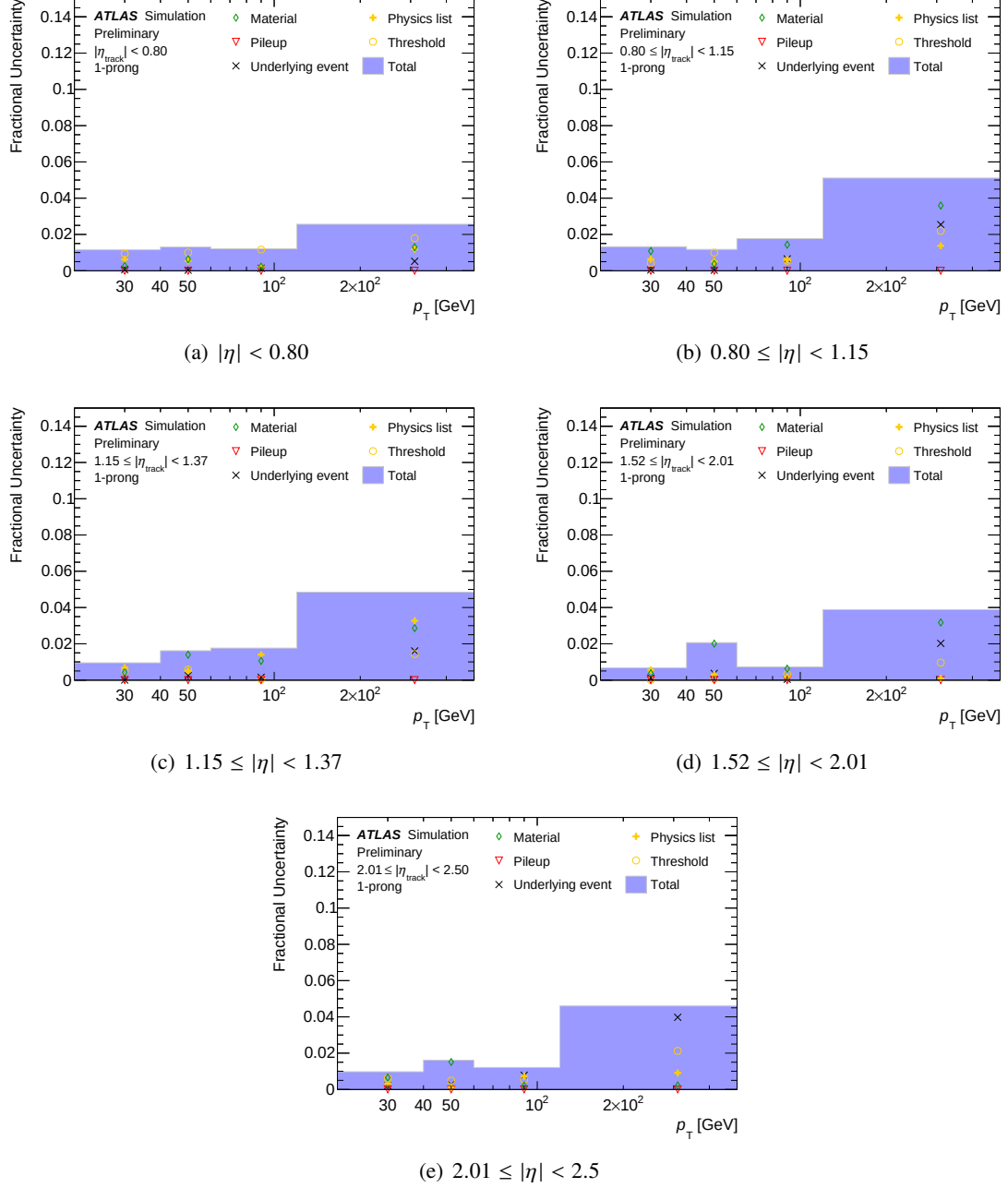


Figure 15: Relative systematic uncertainties on the efficiency of the τ - e discrimination for 1-track $\tau_{\text{had-vis}}$ candidate as a function of the candidate p_T for different η ranges of the track direction. The different sources of systematic uncertainties are listed in the legend, the grey area is the square sum of the individual contribution.

The linear coefficient A corresponding to the average contribution per vertex to the tau energy is shown in Fig. 16. It is computed in bins of $|\eta|$, n_p , $E_{\text{true}}^{\text{vis}}$, and is then averaged over $E_{\text{true}}^{\text{vis}}$ bins. Following the Run-1 convention, the pile-up offset vanishes for the average pile-up condition of the MC sample, i.e. for $N_{\text{PV}} = \langle N_{\text{PV}} \rangle \simeq 14$.

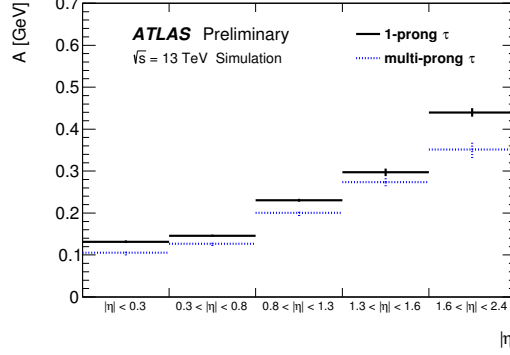


Figure 16: Pileup correction factor $A(|\eta|, n_p)$

The detector response $\mathcal{R}$ is extracted as the Gaussian mean of the $(E_{\text{LC}} - E_{\text{pileup}}) / E_{\text{true}}^{\text{vis}}$ distribution. The response is computed for 1-prong and multi-prong tau decays separately, for various $E_{\text{true}}^{\text{vis}}$ and $|\eta|$ bins. For each pseudorapidity bin, an analytic function is then used to fit the response as a function of the average pileup-corrected energy. The result is displayed in Fig. 17. The rapid drop of the response at low energy is caused by the increasing relative contribution from pileup (given the convention that was chosen for the pileup correction). In the low-energy regime where $E_{\text{T,true}}^{\text{vis}} \lesssim 10$ GeV, the response is extrapolated using a constant function.

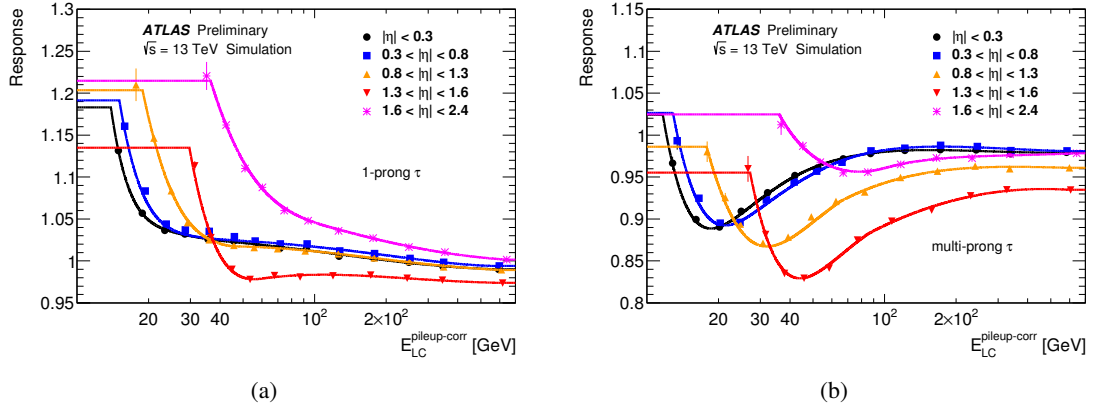


Figure 17: Detector response $\mathcal{R}$ for 1-prong (a) and multi-prong (b) tau decays.

6.2 Resolution and systematics

The resolution on the calibrated energy is computed as the Gaussian width of the $E_{\text{calib}} / E_{\text{true}}^{\text{vis}}$ distribution. As for the response, the resolution is computed for 1-prong and multi-prong tau decays, for several $E_{\text{true}}^{\text{vis}}$ and $|\eta|$ bins. The Gaussian resolutions σ_E , shown in Fig. 18 as a function of the calibrated energy, follow

the usual parameterization

$$\frac{\sigma_E}{E} = \frac{a}{\sqrt{E}} \oplus \frac{b}{E} \oplus c, \quad (3)$$

where a , b , c are the sampling (or stochastic) term, the noise term, and constant term, respectively.

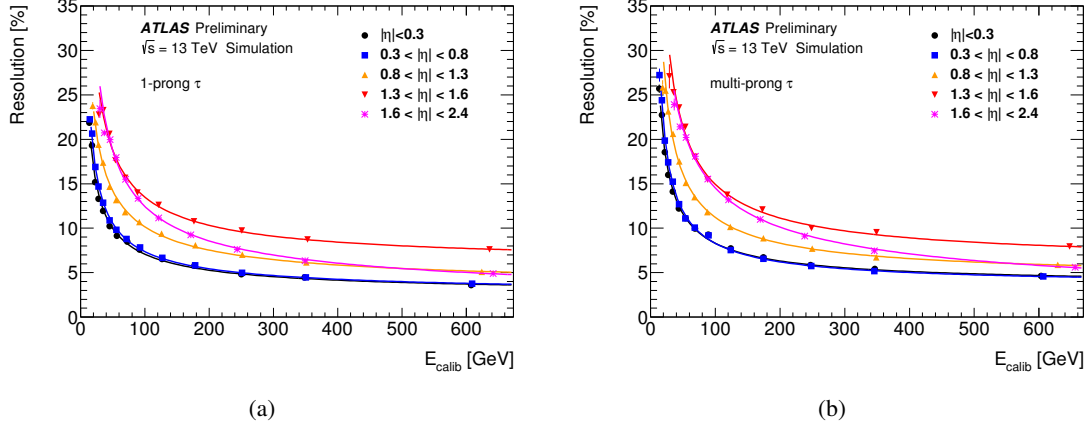
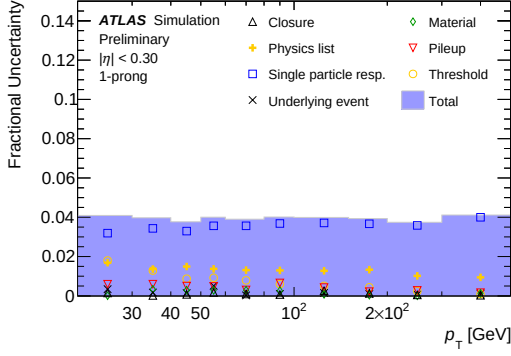


Figure 18: Resolution on the $\tau_{\text{had-vis}}$ energy after calibration for reconstructed 1-prong (a) and multi-prong (b) tau decays. Fits based on the parameterization from equation 3 are overlaid.

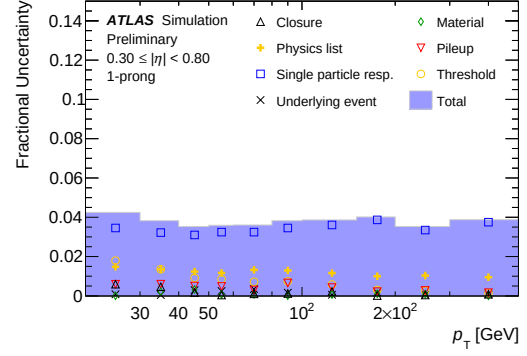
Systematic uncertainties on the tau energy scale calibration are evaluated from simulation, using the alternative MC samples listed in section 3. Systematics are computed as the shift in the Gaussian mean of the $E_{T,\text{calib}}/E_{T,\text{true}}^{\text{vis}}$ distribution of the alternative sample from them of the nominal sample. The non-closure of the calibration in the nominal sample, which corresponds to the deviation from unity of that same Gaussian mean, is also taken into account. The pileup uncertainty, which accounts for the remaining dependence of the calibrated energy on the average number of interactions per bunch crossing μ , is found to be less than 1%. The residual dependence with respect to the number of primary vertices N_{PV} is of the same order. The underlying event tune has a negligible impact on the calibration. The uncertainty on the response is evaluated using the alternative Geant4 hadronic shower model, the alternative material modelling and the variation of the calorimeter noise thresholds. The alternative shower model can affect the clustering of energetic cells into TopoClusters, as well as the classification of TopoClusters into electromagnetic/hadronic categories (which affects the LC-calibrated energy). The amount of material in front of the calorimeter affects both the dead-material correction weights (part of the LC calibration) and the tau response $\mathcal{R}$. The noise threshold variations estimate the sensitivity of the response to the modelling of the calorimeter calibration in the simulation.

A more refined and exhaustive treatment of systematic uncertainties on the response was performed in Run-1 [3]. However, the results of the *particle deconvolution method* [26] (which takes advantage of single-pion response measurements in low- μ data and test beam data) are not yet available for Run-2. While the Run-2 MC-based response uncertainty partly covers the components entering the single-particle response uncertainty, it does not account for the uncertainty on the EM-scale calibration ($\pi^0 \rightarrow \gamma\gamma$). As the detector has not dramatically changed between Run-1 and Run-2, the Run-1 single-particle response uncertainty has been included as an additional systematic uncertainty on the Run-2 tau energy scale, inflated by a safety factor of 2. For 1-prong tau decays with $|\eta| < 0.8$, the single-particle response uncertainty is valid up to $p_T \approx 500$ GeV. Beyond this, test beam data can no longer be used to constrain the charged-pion response.

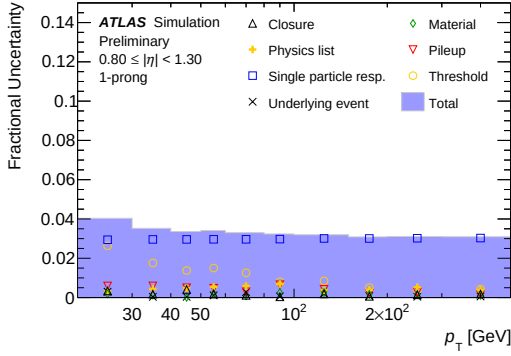
The total uncertainty is obtained by symmetrizing each systematic uncertainty and summing them in quadrature. The result is displayed in Figs. 19 and 20 as a function of the calibrated transverse energy. Uncertainties vary in the 2 – 6% range, depending on p_T and $|\eta|$. As expected the single particle response is always an important component, but at low p_T additional contribution from the calorimeter calibration becomes relevant. In the most central regions, the GEANT4 hadronic shower model contributes significantly to the total systematic uncertainty.



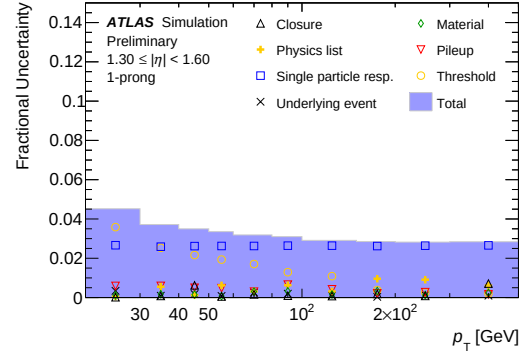
(a) 1-prong, $|\eta| < 0.3$



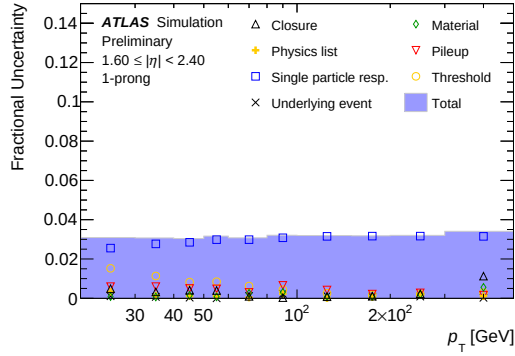
(b) 1-prong, $0.30 < |\eta| < 0.8$



(c) 1-prong, $0.8 < |\eta| < 1.3$

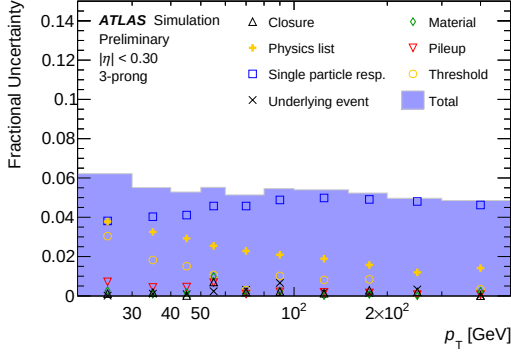


(d) 1-prong, $1.3 < |\eta| < 1.6$

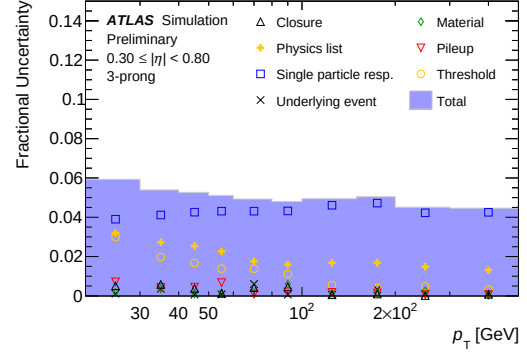


(e) 1-prong, $1.6 < |\eta| < 2.4$

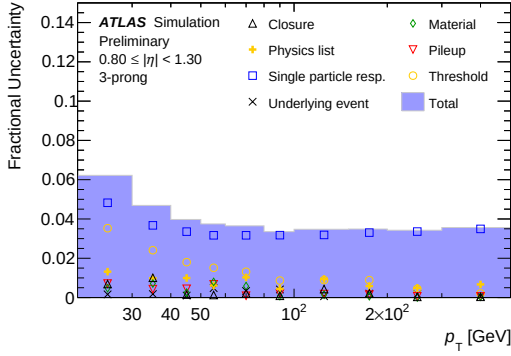
Figure 19: Systematic uncertainties on the $\tau_{\text{had-vis}}$ transverse energy after calibration for 1-prong tau decays. The individual uncertainties as well as their quadratic sum are shown as function of the $\tau_{\text{had-vis}}$ p_T for different $|\eta|$ ranges.



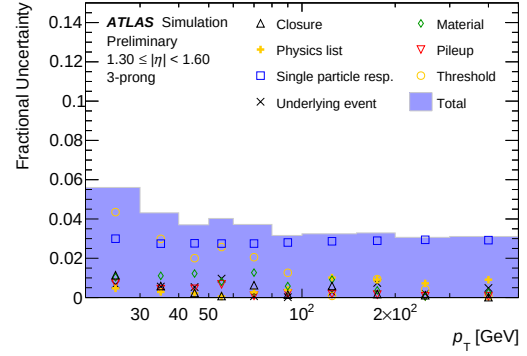
(a) 3-prong, $|\eta| < 0.3$



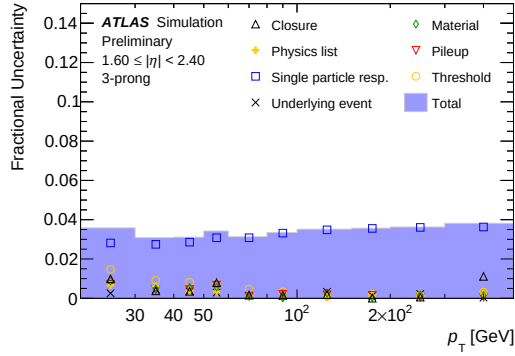
(b) 3-prong, $0.30 \leq |\eta| < 0.80$



(c) 3-prong, $0.80 \leq |\eta| < 1.30$



(d) 3-prong, $1.30 \leq |\eta| < 1.60$



(e) 3-prong, $1.60 \leq |\eta| < 2.40$

Figure 20: Systematic uncertainties on the $\tau_{\text{had-vis}}$ transverse energy after calibration for 3-prong tau decays. The individual uncertainties as well as their quadratic sum are shown as function of the $\tau_{\text{had-vis}}$ p_T for different $|\eta|$ ranges.

7 Conclusion

The reconstruction, identification and energy calibration of hadronically decaying tau leptons has been updated for the LHC Run-2 data taking conditions with centre-of-mass energy $\sqrt{s} = 13$ TeV and 25 ns bunch spacing.

The algorithms have been tuned on simulated samples and provide three nominal identification operation points with increasing background rejection. The combined reconstruction and identification efficiencies for these operating points are 0.6 (0.5), 0.55 (0.4) and 0.45 (0.3) for 1-prong (3-prong) tau decays. Rejection against electrons is achieved using a special working point of the offline electron identification tool, providing 95% efficiency and an inverse background efficiency between 20 and 200, depending on the pseudorapidity of the $\tau_{\text{had-vis}}$ candidate.

Systematic uncertainties have been evaluated before the start of data-taking using simulated samples corresponding to changes to the underlying event tune, the simulation of the hadronic shower in the calorimeter, a varied model of the ATLAS geometry and material distribution, and a variation in the noise thresholds of the calorimeter clustering procedure. They depend on the $|\eta|$, p_T and charged particle multiplicity of the $\tau_{\text{had-vis}}$ candidates. They are in the range 2–4.5% for reconstruction efficiency, 5–18% for the identification efficiency, 1–5% for electron rejection and 3–6% for the energy scale.

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